

Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arching, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



⚠ WARNING ⚠

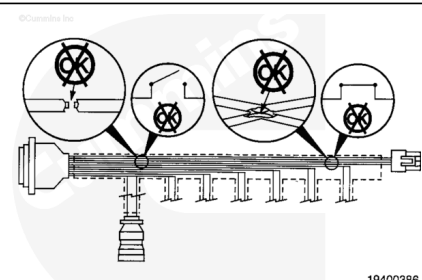
Wear safety glasses or a face shield, as well as protective clothing, to prevent personal injury when using a steam cleaner or high-pressure water.

Prepare engine to remove the wire harness.

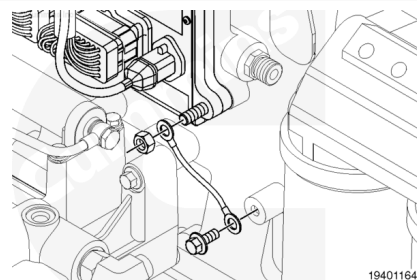
- Disconnect the battery. Refer to Procedure 013-009 in Section 13 of the QSK19 Service Manual, Bulletin 4021592. (</qs3/pubsys2/xml/en/procedures/20/20-013-009.html>)
- Clean engine at all electrical connections. Refer to Procedure 000-009 in Section 0 of the QSK19 Service Manual, Bulletin 4021592. (</qs3/pubsys2/xml/en/procedures/99/99-000-009.html>)

Remove

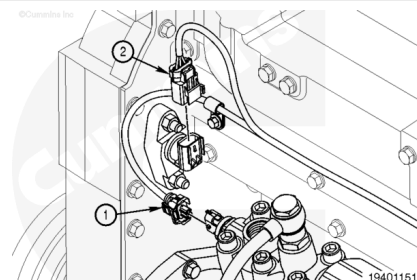
Replace the engine harness if there is an open or short circuit found under the protective covering of the harness body.



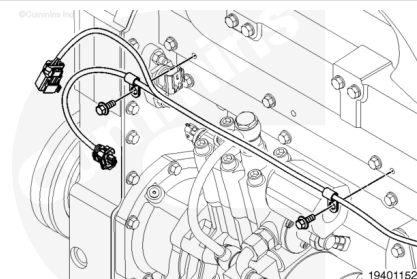
Disconnect the engine block ground from the engine.



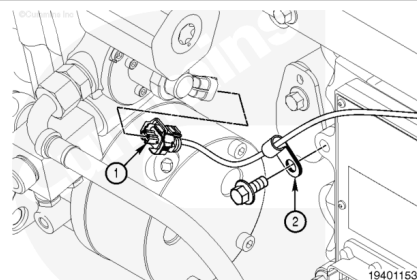
Disconnect the injector metering rail 1 pressure sensor (1) and engine camshaft speed/position sensor (2).



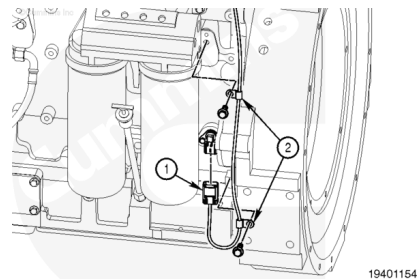
Remove the wire harness clamps.



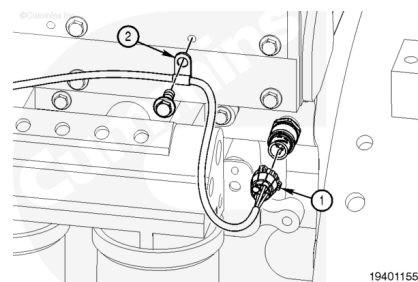
Disconnect the fuel pump pressurizing assembly 1 (1) and remove the wire harness clamps (2).



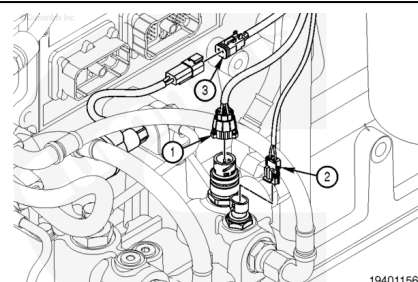
Disconnect the engine crankshaft speed/position sensor (1) and remove the wire harness clamps (2).



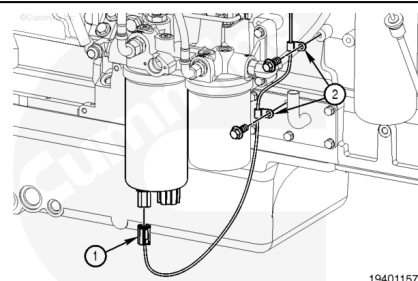
Disconnect the engine oil rifle pressure 1 sensor (1) and remove the wire harness clamps (2).



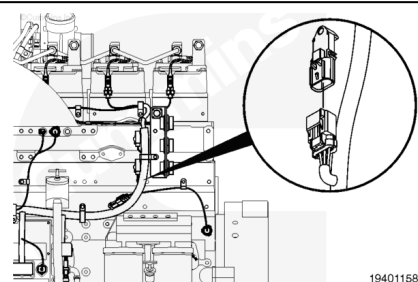
Disconnect the fuel delivery pressure sensor (1) and the engine fuel temperature sensor 1 (2). Remove the electric lift pump for engine fuel supply (3).



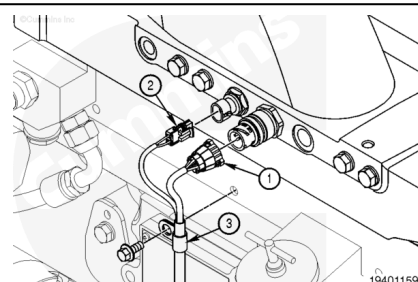
Disconnect the water-in-fuel indicator sensor (1) and remove the wire harness clamps (2).



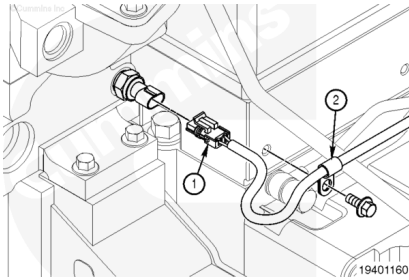
Disconnect the barometric pressure sensor.



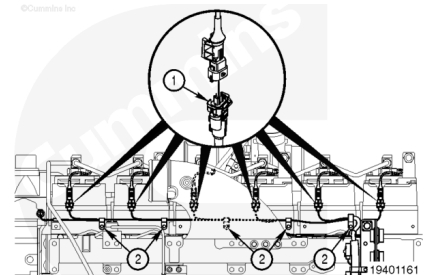
Disconnect the intake manifold 1 pressure sensor (1) and intake manifold 1 temperature sensor (2), and remove the wire harness clamps (3).



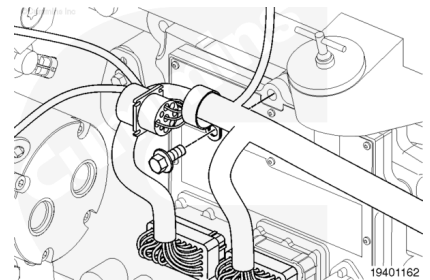
Disconnect the engine coolant temperature 1 sensor (1) and remove the wire harness clamps (2).



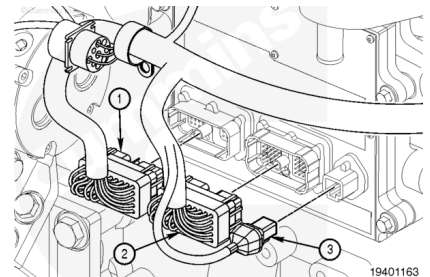
Disconnect the injector harness connectors (1) and remove the wire harness clamps (2).



Remove the wire harness clamp from the top of the engine control module (ECM).



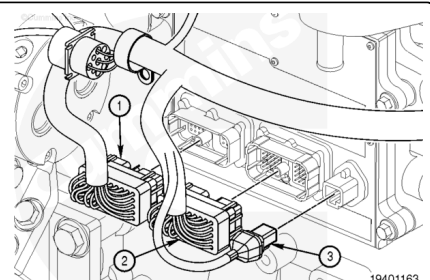
Disconnect the ECM 60-pin original equipment manufacturer (OEM) port connector (1), the ECM 60-pin engine port connector (2), and the ECM 4-pin power connector (3).



Install

Connect the ECM 60-pin OEM port connector (1), the ECM 60-pin engine port connector (2), and the ECM 4-pin power connector (3).

Use an inch-pound torque wrench, Part Number 3376592, with 4 mm [5/32 in] hex head adapter to tighten the connector jackscrew.



Torque Value: 2.8 n•m [25 in-lb]

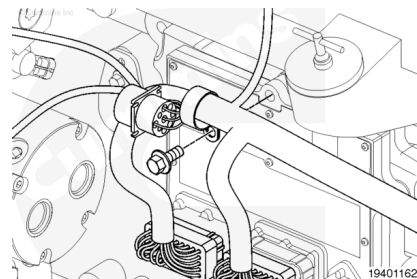
Note : Do not over-torque as connector damage can occur.

Attach the wire harness clamp to the top of the ECM.

Torque Value:

Wire Harness Clamp

1.40 n•m [30 ft-lb]

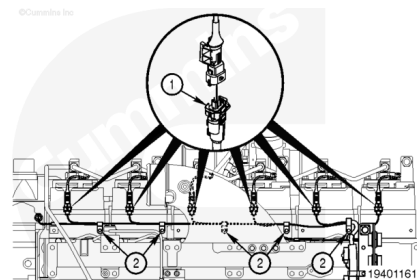


Connect the injector harness connectors (1) and attach the wire harness clamps (2).

Torque Value:

Wire Harness Clamp

1.40 n•m [30 ft-lb]

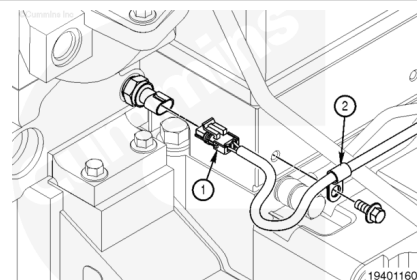


Connect the engine coolant temperature 1 sensor (1) and attach the wire harness clamps (2).

Torque Value:

Wire Harness Clamp

1.40 n•m [30 ft-lb]

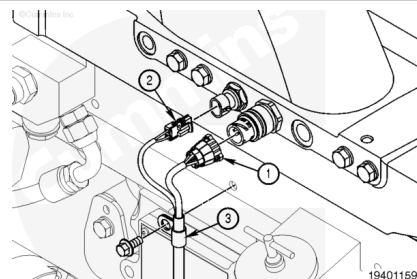


Connect the intake manifold 1 pressure sensor (1), intake manifold 1 temperature sensor (2), and attach the wire harness clamps (3).

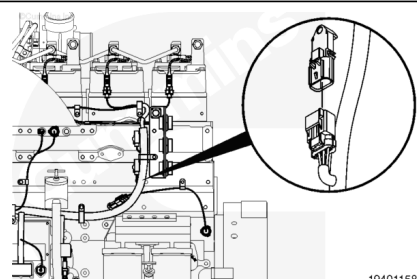
Torque Value:

Wire Harness Clamp

1.40 n•m [30 ft-lb]

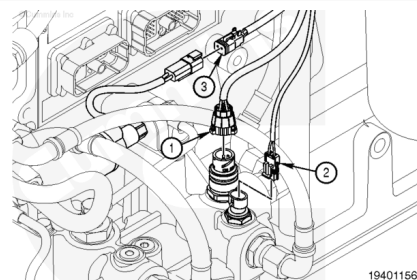


Connect the barometric pressure sensor.



Connect the fuel delivery pressure sensor (1) and the engine fuel temperature sensor 1 (2).

Remove the electric lift pump for engine fuel supply (3).

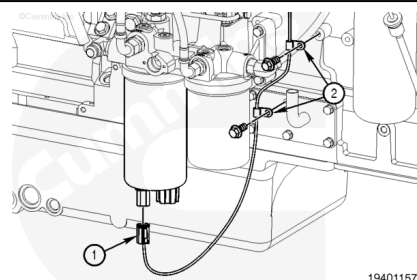


Connect the water-in-fuel indicator sensor (1) and attach the wire harness clamps (2).

Torque Value:

Wire Harness Clamp

1. 40 n•m [30 ft-lb]

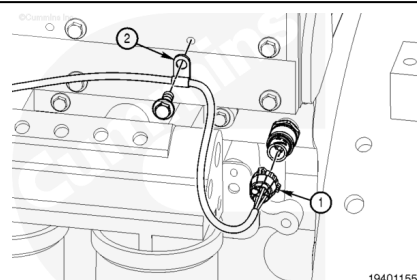


Connect the engine oil rifle pressure 1 sensor (1) and attach the wire harness clamps (2).

Torque Value:

Wire Harness Clamp

1. 40 n•m [30 ft-lb]

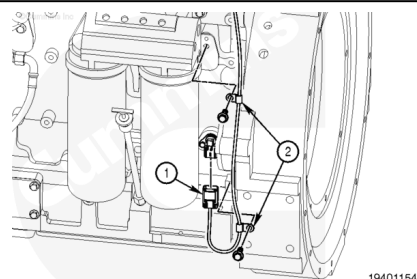


Connect the engine crankshaft speed/position sensor (1) and attach the wire harness clamps (2).

Torque Value:

Wire Harness Clamp

1. 40 n•m [30 ft-lb]

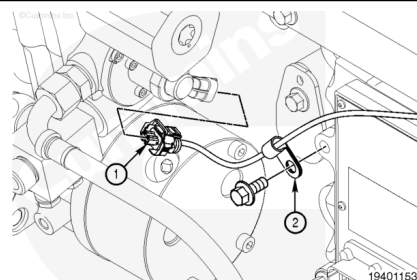


Connect the fuel pump pressurizing assembly 1 (1) and attach the wire harness clamps (2).

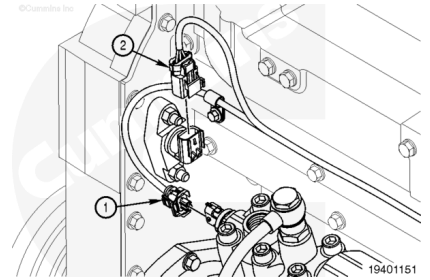
Torque Value:

Wire Harness Clamp

1. 40 n•m [30 ft-lb]



Connect the fuel pump pressurizing assembly 1 (1) and engine camshaft speed/position sensor (2).

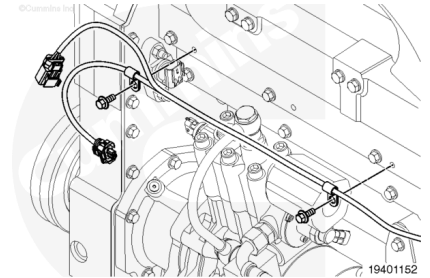


Attach the wire harness clamps.

Torque Value:

Wire Harness Clamp

1.40 n•m [30 ft-lb]

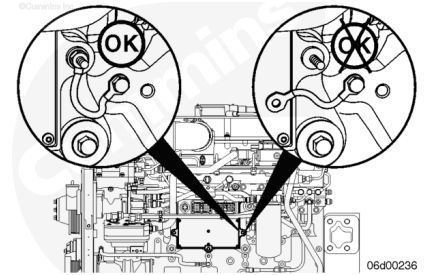


Connect the engine block ground from the engine.

Make sure the ground strap is properly installed. A missing or poorly connected ground strap can cause intermittent engine performance problems.

The ground strap **must** be connected between the head of the ECM mounting bolt and the ECM.

The ground strap **must** be connected to an unpainted surface that is free of corrosion.



Torque Value:

Wire Harness Clamp

1.40 n•m [30 ft-lb]

Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



- Connect the battery. Refer to Procedure 013-009 in Section 13 of the QSK19 Service Manual, Bulletin 4021592. (</qs3/pubsys2/xml/en/procedures/20/20-013-009.html>)
- Operate the engine and check for active fault codes.

Last Modified: 17-Apr-2014
